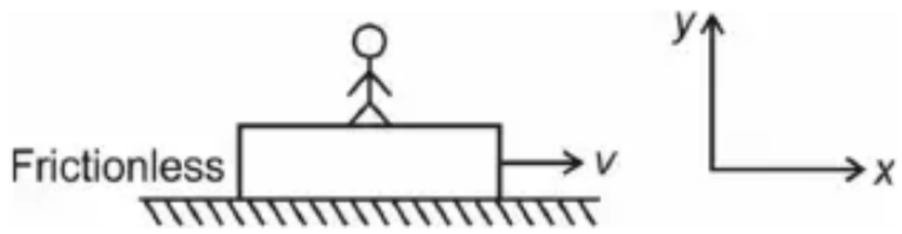
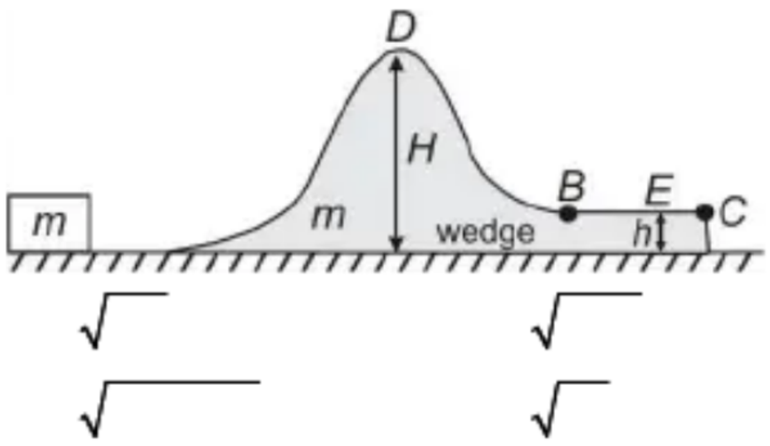
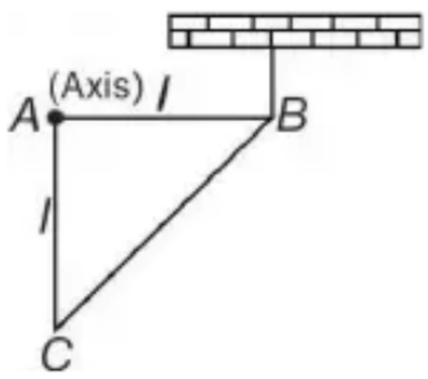
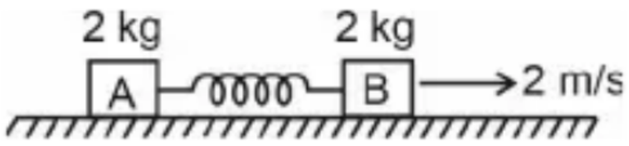
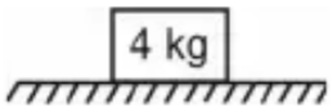

v
15 kg
30 kg
2 m/s


Frictionless
v
y
x


D
H
m
m
wedge
B
E
h
C


(Axis)
A
l
B
l
C


2 kg
2 kg
A
B
2 m/s


4 kg


10 m/s
5 m/s

ZOOLOGY

Concept: Human Respiratory System

40. Read the following statements and select the **correct** option.

Statement X: The trachea, primary, secondary, tertiary bronchi, initial bronchioles and terminal bronchioles are supported by incomplete cartilaginous rings.

Statement Y: During swallowing, epiglottis can be covered by a thin elastic cartilaginous flap called glottis to prevent the entry of food into the sound box.

(1) Both statements X and Y are correct

(2) Only statement X is correct

(3) Only statement Y is correct

(4) Both statements X and Y are incorrect

Concept: Human Respiratory System

41. Select the **correct** statement w.r.t human beings.

(1) Lungs are covered by a double layered peritoneum

(2) Lungs are placed above the diaphragm

(3) The thoracic chamber is formed dorsally by sternum and ventrally by the vertebral column

(4) Cellular respiration occurs at alveolar level

Concept: Formed Elements

42. Select the blood cell whose increase in number is associated with the exaggerated response of the immune system to certain antigens present in the environment.

(1) Lymphocytes (2) Neutrophils

(3) Monocytes (4) Eosinophils

Concept: Cardiac Cycle

43. How many of the following given below in the box have clinical diagnostic significance?

Dub, BP, ECG, VC, Urine, Lub

(1) Five (2) Six

(3) Three (4) Four

Concept: Mechanism of Breathing

44. Select the option that **correctly** mentions the muscle movements associated with forceful expiration.

(1) Diaphragm relaxes and external inter-costal muscles contract

(2) Internal inter-costal muscles contract and abdominal muscles relax

(3) External inter-costal muscles relax and diaphragm contracts

(4) Abdominal muscles contract along with internal inter-costal muscles

Concept: Glomerular Filtration

45. Read the statements w.r.t. urine formation.

(A) Around 1100-1200 mL of blood is filtered by the kidneys per minute.

(B) Podocytes are arranged in an intricate manner so as to leave minute spaces called slit pores.

(C) Blood is filtered finely through slit pores and all the constituents of plasma pass into the lumen of Bowman's capsule.

(D) An adult human excretes around 1-1.5 litre of urine per day.

Choose the option that **correctly** states the above statements as true (**T**) or false (**F**).

	A	B	C	D
(1)	T	F	F	T
(2)	F	T	F	T
(3)	T	F	T	T
(4)	T	T	F	T

Concept: Types of Nephron

46. Vasa recta is absent or highly reduced in

(1) Cortical nephrons

(2) Juxtamedullary nephrons

(3) Cortical and juxtamedullary nephrons

(4) Juxtamedullary nephrons of aged persons only

Concept: Exchange of Gases

47. Choose the option with only **incorrect** statement(s).

a. Respiratory diffusion membrane in humans is avascular and moist.

b. Respiration helps in elimination of carbon dioxide and water.

c. Larynx provides passage to both food and air.

(1) a and c (2) b and c

(3) a, b and c (4) Only c

Concept: Cardiac Cycle

48. Stroke volume of an adult human heart is represented as

(1) Heart beats per minute × Cardiac output

(2) Cardiac output ÷ Heart beats per minute

(3) Volume of left ventricle ÷ Heart beats per minute

(4) Heart beats per minute × Volume of left ventricle

Concept: Electrocardiogram(ECG)

49. Which of the following cardiac events results immediately after the QRS wave complex in a standard ECG?

(1) The tricuspid and bicuspid valves open

(2) Ventricular systole

(3) Blood from vena cava is forced into the left atrium

(4) The ventricles relax and the semilunar valves are closed

Concept: Functions of Tubules

50. Select the **incorrect** statement w.r.t. PCT of nephron.

(1) It reabsorbs 70-80% of electrolytes and water

(2) Lining of simple ciliated epithelium increases the surface area for reabsorption

(3) Helps to maintain the pH and ionic balance of body fluids

(4) Selectively secretes hydrogen ions and ammonia

❑ ❑ ❑